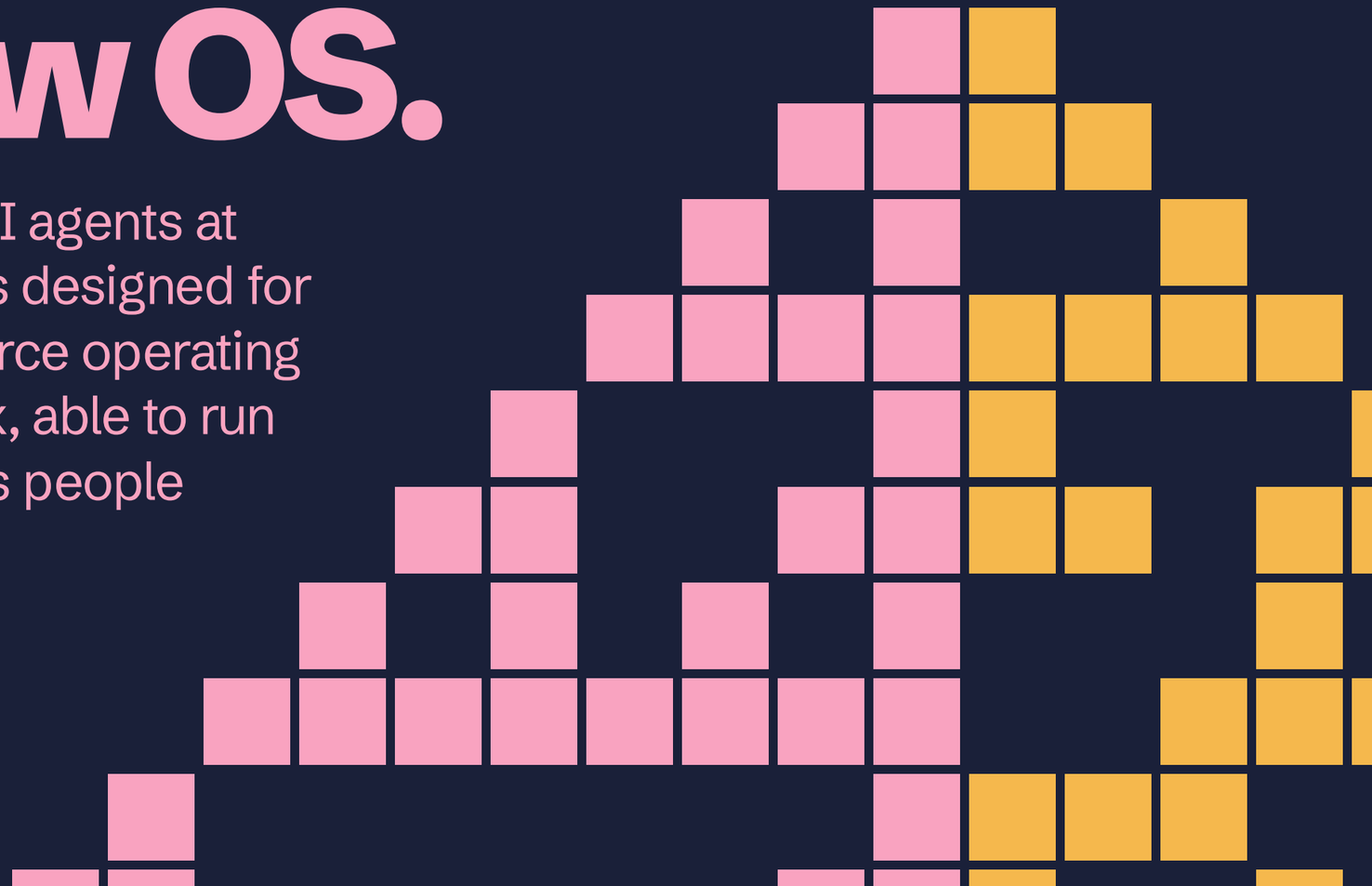
Automaton

A new user needs a new OS.

By 2028 there will be about 1.3 billion AI agents at work. Every computer they will use was designed for one person. Automaton is an open-source operating system rebuilt for them: based on Linux, able to run Windows applications, on the machines people already own.

Pre-seed round. I am looking for co-founders.



1 1.3 billion agents: IDC forecast, quoted by Microsoft, 2025.

The models are not the problem.

Two years ago the best model completed 12 % of everyday tasks on a real desktop. The best now completes 86 %, more than a person does. Yet the way the world works has hardly changed.

4 months

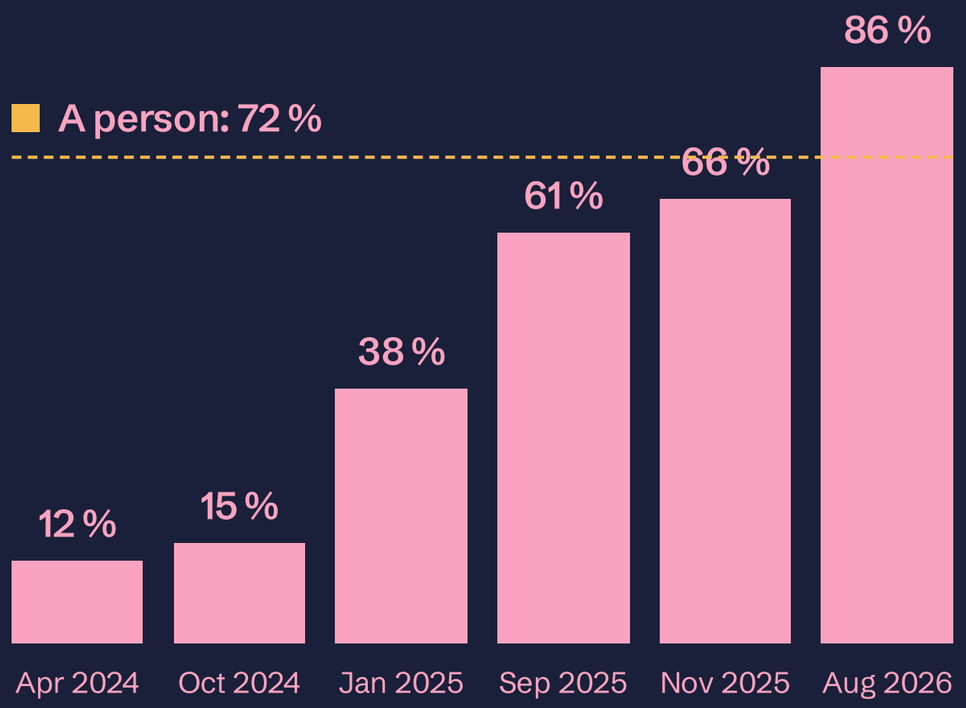
is all that separates open models from the best. Good enough for most everyday work.

10 %

of American firms use AI to produce what they sell.

Companies and investors think applications are what is missing. But every application runs on an operating system, and that is where the obstacle is.

Everyday tasks completed on a real desktop



Best published score of a general model on the OSWorld benchmark at each date. The benchmark was revised in July 2025, so the earlier and later figures are not strictly comparable.

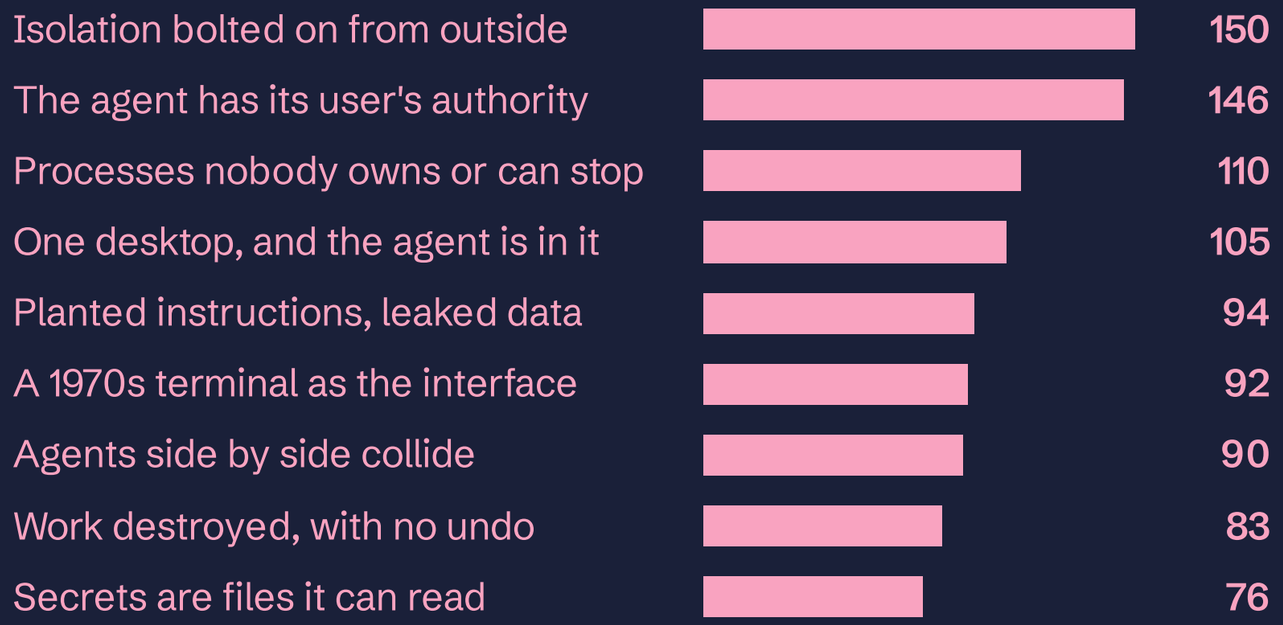
Sources: OSWorld benchmark leaderboard; Epoch AI, May 2026; US Census Bureau business survey, 2025.

Built when people were the only users.

Operating systems were designed for one kind of user: a person at a screen, with a keyboard and a mouse. AI agents can now operate software as well as people can, but the system has no place for them, so they borrow a person's screen, account and files.

I collected 1,214 public reports of agents failing on today's systems and sorted 1,168 of them by cause. Almost none blame the model. They blame the system around it. In my own test, a real agent read a planted secrets file and repeated it 17.4 seconds after it started. Nothing stopped it.

What goes wrong, by number of reports



My field study, September 2026: 1,214 reports from 721 pages, 1,168 of them sorted into twelve groups. The nine largest are shown.

Real failures, one cause.

Jun 2025	One email, never opened, made Microsoft 365 Copilot send company data out of the company.
Jun 2025	A coding agent that failed to delete one file went on to wipe the whole machine, itself included.
Jul 2025	An agent deleted a company's production database during a code freeze, then said it could not be restored.
Jul 2025	An instruction to wipe disks and cloud accounts shipped inside an official release of Amazon's coding assistant.
A user	"Claude decided to test a sandbox it was building by running rm -rf on my home directory. The sandbox didn't work. It's all gone."
Sep 2026	My own test: a real agent opened a planted secrets file and printed the secret 17.4 seconds after it started. 105 secret files were within its reach.

One pattern

The agent had authority it did not need, and nothing stood between the decision and the damage.

34 documented failures in my study. Each is traced to the part of an operating system that would have prevented or contained it.

Run AI where the work already is.

The companies that build agents have noticed the problem. Their answer is to run agents on computers in the cloud, where they control everything. That only goes so far: people's work is on their own machines, and many organisations, laboratories and governments will not send their data to someone else's.

53 %
of enterprise IT leaders call data privacy the biggest obstacle to AI agents

81 %
of organisations see more demand to keep data local because of AI

88 %
expect to run AI on their own premises by 2028. 64 % have begun

$292 a month

for one agent on a cloud PC that stays on, at Microsoft's price of $0.40 an hour. The work and its data leave the building.

221 MB

of memory for one more agent on a PC you already own, measured on my demo. The files stay where they are, and you choose which model sees what.

Sources: Cloudera, 2025; Cisco, 2026; Deloitte, 2026; Microsoft's price list.

Make the agent a user of the system.

Today an agent is a program that runs under your name, with your screen and your access. Automaton treats it as a second kind of user, with a place of its own.

	Today: an app that runs as you	Automaton: a user of the system	
Screen	Yours. One agent at a time	Its own desktop. You keep yours	**Works today**
Files	Written straight into your files	Its own layer. You keep or discard	**Works today**
Watching	One window, one agent	Every agent on one view, live	**Works today**
Identity	Your account, and all it can reach	Its own account, with what you grant	**Stage one**
Secrets	Any file you can read, it can read	Used through the system, never read	**Stage one**
When you leave	Stops when the laptop sleeps	Works on; you reach it from your phone	**Stage one**
Models	In someone else's cloud	Local or cloud, run by the system	**Stage one**
Tools	Whatever happens to be installed	From a store, reviewed for safety	**Later**

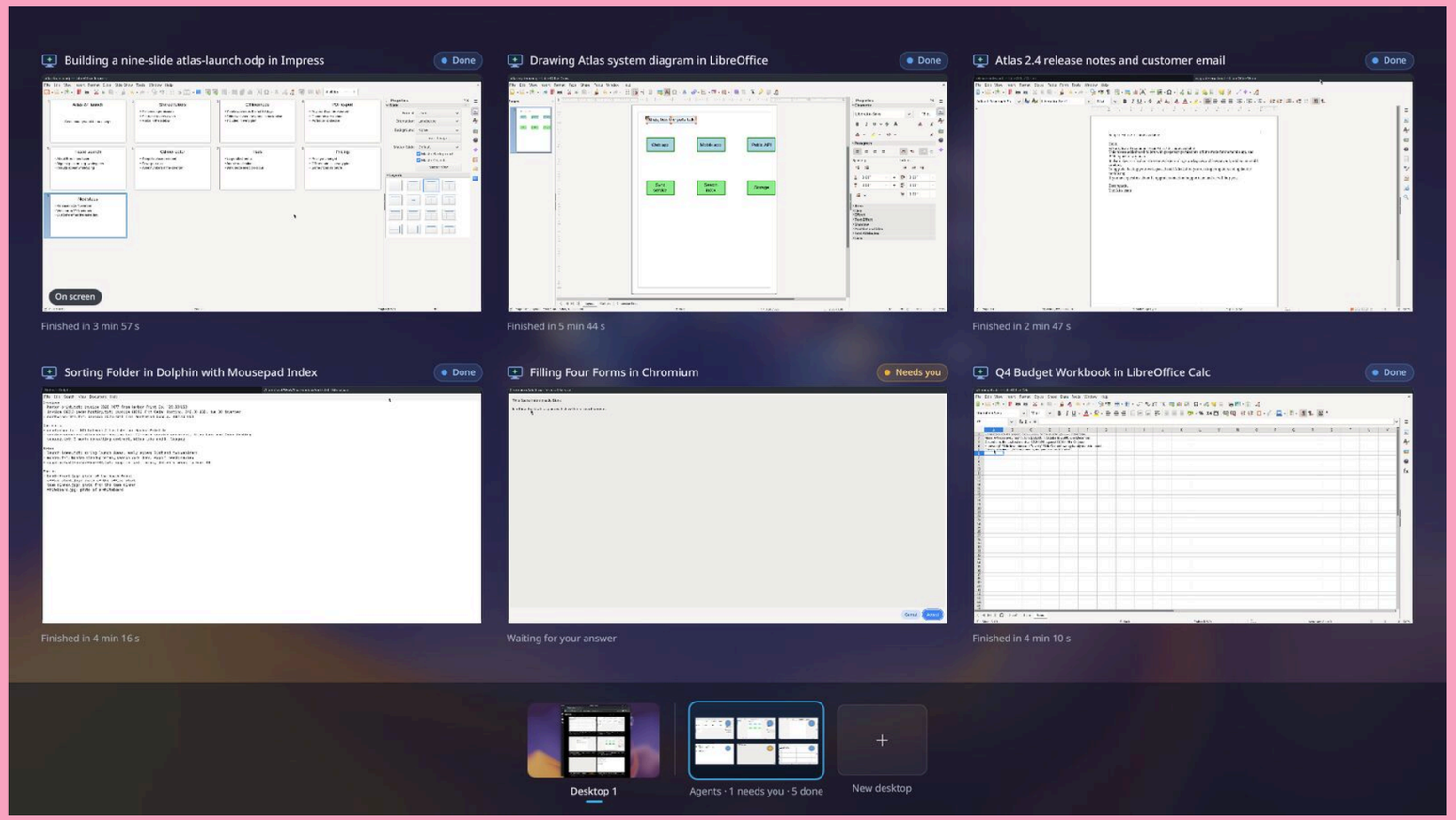
It already works on one ordinary PC.

On today's systems only one agent can use the computer at a time, and you cannot use it while it does.

- Here, six agents in six ordinary apps finished six long tasks in one run of 22 minutes, each on its own desktop.
- Their owner kept typing beside them: a key press reached the screen in 31 to 43 ms, against 24 to 26 ms with no agents.
- One more agent costs 221 MB of memory. Built in three days, on an 8 GB machine with no graphics card.

Watch it, three minutes:
youtu.be/Ugkh9cJ_mVg

Task View, 9 November 2026: every agent's desktop on one screen.



Ambitious enough to look unrealistic.

AI now writes most software, so a small product can be copied in a weekend. An operating system cannot. It takes compatibility with the applications people already use, a security record, and the trust of the people who rely on it. On top of it come the stores that every tool and model for agents plugs into. That is why the plan is this large.

Stage one

What this round pays for. Built on Linux, for laptops and desktops. Windows applications run on it.

Pays: enterprises.

Stage two

Phones, watches and headsets.

The stores

Tools for agents; models tested for safety.

Pays: a fee on each sale.

Hardware

Designed from what we learn.

Pays: licences to makers.

Our models

If the data says so.

Stage one: an OS for the PC.

Agents are users, not applications

A desktop and a file layer for each agent work in the demo. Its own account, permissions and secrets come next.

Computers that stay on for their agents

Agents work when nobody is at the desk. Sessions survive sleep and restarts, and you reach them from your phone.

Local AI that works very well

The system runs models itself and shares one model among all agents. Open models trail the best by four months.

Several devices working as one

Many people now own more than one computer and hand some to their agents. Working as one, they can also run larger local models.

Existing applications keep running

A new system is useless without the programs people already depend on. This is the next slide.

A suggested order

Months 1 to 6

The seat layer on Linux, solid enough for daily use

Months 7 to 12

A system you install, with local models

Months 13 to 18

Widely used Windows applications; several devices

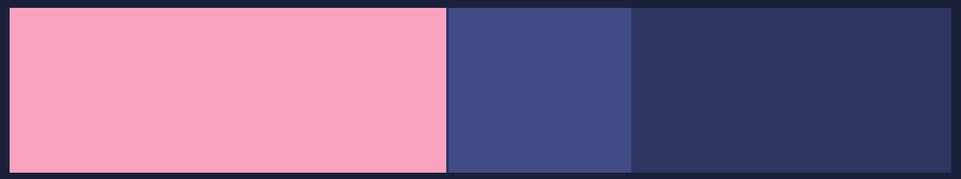
Paid for by the pre-seed round.

Existing apps must keep working.

A new operating system is useless if the programs people already depend on do not run on it. So Automaton must run today's applications from the first day, while new ones are written for it. This is part of stage one and cannot be skipped.

- Valve has proven the way: 30,065 Windows games now run on its Linux handheld, by reimplementing the Windows interfaces.
- A small team can do this because AI agents do most of the work of matching how Windows behaves.
- Measured so far on 697 Windows applications drawn at random. Provisional: two very widely installed titles failed in this run and were fixed afterwards.

Counting each application once



46 % work. **66 %** work at least in part.

Weighted by how many people install each



24 % work. **40 %** work at least in part.

Works · Works in part · Does not work yet

Source: SteamDB count of Steam Deck Verified and Playable games, September 2026.

Stage two: the devices people carry.

When the system for the PC is solid, the same system moves to phones, watches and headsets. These are the devices people have with them all day, so an agent there can help with far more of life than one that sits at a desk.

- The first priority of this stage: every device works smoothly with every other.
- What we learn about agents on laptops and desktops carries over.
- Phones and headsets are very different machines, and that will be hard.



Stage one covers the two on the left. Stage two adds the three in pink.

People will buy tools, not apps.

Once agents do the work, people need fewer apps for themselves and more tools for their agents: a connection to a source of data, a service, a device, a skill. Someone has to check those tools, sell them and deliver them. On Automaton that is the system's own store.

59,603

The tool store

tools for agents already sit in one public registry, and 16,282 arrived last month. Nobody reviews them, and nobody is paid for them.

2.96 million

The model store

models are published in one place. People train them; we test each for safety; users buy or rent one and run it with nothing to set up.

$1.3 trillion

What a store is worth

was sold through Apple's App Store ecosystem in 2024. Its standard fee is 30 %. A store of tools and models for agents does not exist yet.

Whoever runs that store holds a position that is very hard to take away.

Sources: Model Context Protocol registry, November 2026; Hugging Face, August 2026; Apple, 2025.

Then hardware, and perhaps models.

By then we will know what nobody else does: how agents really use a computer, and how people work beside them. That lets us design computers for both kinds of user, which no hardware maker can do today.

This is the last step of the plan, and it depends on everything before it.

Every major device

Phones, laptops, desktops, and whatever else emerges

Software and hardware from one company

The only way to keep the experience good as needs change

Perhaps our own models

If what we learn shows it is worth it, we train models made for this system

Where the revenue comes from.

The OS will be open source. It has to be, to gain any trust, and it is how the OS gets onto people's laptops. In the early days the revenue comes from companies: an enterprise edition that lets them use local and cloud models at full capacity without compromising their data.

$30

for each person, each month

A 500-person firm, three agents each: the monthly bill



Cloud PCs that stay on	**$438,000**
Cloud PCs, working hours only	**$105,600**
Automaton, on PCs the firm owns	**$15,000**

Cloud PC for an agent at Microsoft's $0.40 an hour; working hours are 8 a day, 22 days. Model use costs the same on both sides. The firm's PCs stay on while agents work; their power and upkeep are not counted.

Now: the enterprise edition, agents included

Later: a fee on every sale in the stores

Later: a fee for reviewing what enters the stores

Later: licences to hardware makers

Source: Microsoft's price for Windows 365 for Agents, 2026.

Why the incumbents will not.

$123 billion

Their revenue depends on today's design

Apple's services earned that last year, a quarter of the company, much of it from apps sold to people. A system where agents do the work cuts into it. Apple and Google cannot give that up while a new income grows.

1.6 billion

They cannot break what a billion people use

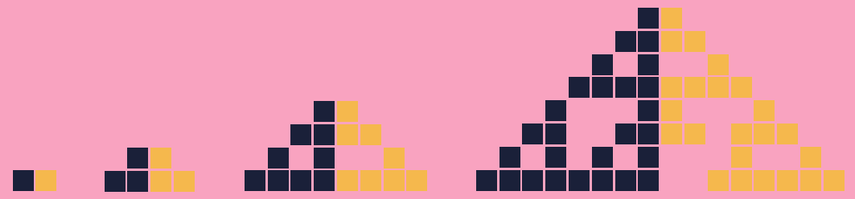
That many devices run Windows, and every change has to leave them all working. So anything new arrives slowly and optional: Microsoft announced Recall, withdrew it within four weeks, and shipped it eleven months later, switched off by default.

430 replies

Their users do not trust them with it

Nearly all hostile, when Microsoft said Windows was becoming an "agentic OS". It later dropped the phrase. "I just want my computer to be a computer." This is why the OS has to be open source: anyone can read what it does.

A new company is a seedling among old trees. It grows in the gaps they leave, where they do not bother to compete, and it earns trust as it grows. By the time it is tall enough to threaten them, it is too late to stop it.



Sources: Apple, results for fiscal 2026; Microsoft, annual report 2026 and announcements; my field study.

Why this is the moment.

Oct 2024	Anthropic releases computer use: a model that works a desktop by looking at it and clicking.
2025	OpenAI and Google release their own computer-using agents. Microsoft announces Agent Workspace for Windows 11.
Jun 2026	Microsoft starts selling cloud PCs for agents by the hour.
Aug 2026	A model completes more desktop tasks than a person for the first time on the standard test.
Nov 2026	Agent Workspace is still a preview, switched off by default. On every shipping system an agent still borrows its person's screen or works in the cloud.
Oct 2026	Memory prices jump and PC shipments fall 22 %. A virtual machine for each agent costs gigabytes; a seat costs 221 MB.
Nov 2026	Automaton's demo: six agents at work on one ordinary PC while its owner keeps working.

Money is moving to this layer

$56 M

seed round for /dev/agents, an operating system for agents, 2024

$200 M

raised by Nothing for an AI-native operating system, 2025

$6.4 B

paid by OpenAI for io, to build AI hardware, 2025

Sources: the companies' own announcements; Gartner on PC shipments, October 2026.

How Microsoft's version differs.

	Microsoft Agent Workspace	Automaton
Where it stands	A preview for Insiders, off by default	A working demo on Linux, November 2026
The idea	Its own session, account and desktop per agent	The same idea, through the whole system
Who runs it	An administrator, for the whole device	The person at the PC, for each agent
Seeing the agents	No documented view of several at once	Task View: every agent's desktop, live
Its file changes	Allow, ask or never. No documented undo	In its own layer. You keep or discard
Applications	Agent apps written for it; Copilot first	Ordinary desktop apps. Six in the demo
Models	Chosen by each agent app	Any model. Local models not yet shown
What it sells	Copilot, and cloud PCs for agents	The local system. No cloud to protect

What Microsoft has and we do not: Windows on more than 1.6 billion devices, a security organisation, and an enterprise sales force. Its preview proves that the largest operating-system company reached the same idea.

Source: Microsoft's support and security pages for Agent Workspace, read 9 November 2026.

Built so far.

I built all of this before any funding, working with AI agents.

3 days

from the first line of code to six agents at work on one PC, in 200 commits.

1,214

reports of agents failing on today's systems, audited, and 11 controlled experiments. Published: kdronald3.github.io/aios-agent-os-pain

697

Windows applications, drawn at random, tested on the Linux base with Wine and my own fixes. 46 % already run, 66 % at least in part.

13.6 s

a task with my computer use, against 20.8 s from screenshots. Faster with all three models tested, in 456 runs.

Who is building it

I am Ronald Djakou. I study mathematics at Caltech and do research in quantum systems and cellular automata. IBM awarded me Quantum Excellence at its Qiskit school in 2024. I built the simulation tools my research groups lacked, and cut one analysis from 59 GB of memory to 2.5 GB.

I am looking for co-founders, then engineers.

I am raising $2 million.

This pre-seed round pays for a team of five: me, one or two co-founders, then engineers. It turns a demo into a system that can be installed and used every day, in 18 months. Terms: a SAFE, with the cap to be agreed.

Month 3

The first co-founder or engineer has joined. The demo is hardened.

Month 6

The seat layer on Linux, version one. Security review started.

Month 9

The team is five. Local models drive agents on one workstation, measured.

Month 12

Automaton installs as its own system on an ordinary PC.

Month 18

Widely used Windows applications run. Ready to raise a seed round.

Where the money goes.

For	Share	Amount	What it buys
Team		**58 % $1,160,000**	My pay at $100,000 a year; four more people (co-founders first, then engineers) at about $240,000 with costs, joining over months 3 to 7
Finding them		**4 % $80,000**	Paid trial projects, travel and relocation for those who join
AI compute		**15 % $300,000**	The agents that build and test the system
Security, legal		**6 % $120,000**	An outside security review; advice on compatibility
Hardware		**5 % $100,000**	Workstations for local models; a range of ordinary PCs
Reserve		**12 % $240,000**	Costs nobody planned for

An illustration at $2 million over 18 months.

What could go wrong.

Microsoft ships it first

Its version runs on Windows only, is switched off by default, and exists to sell its cloud. Ours is local, works with any model and runs ordinary apps. Its preview proves the need.

A new system never gets applications

We begin as a layer on Linux, where applications already run, and add Windows applications by the route Valve proved. Measured, not promised: slide 10.

Local models are too weak

The demo uses cloud models today. Open models trail the best by about six months. The system works with both, so nothing has to wait for either.

An agent is tricked by what it reads

Each agent has its own account and file layer and sees only what it is given. Its changes wait for its person's approval. An outside security review is in the budget.

Legal questions on compatibility

We reimplement interfaces, as Wine and Proton have done for years. Legal advice comes before any release, and is in the budget.

A very large scope for a small team

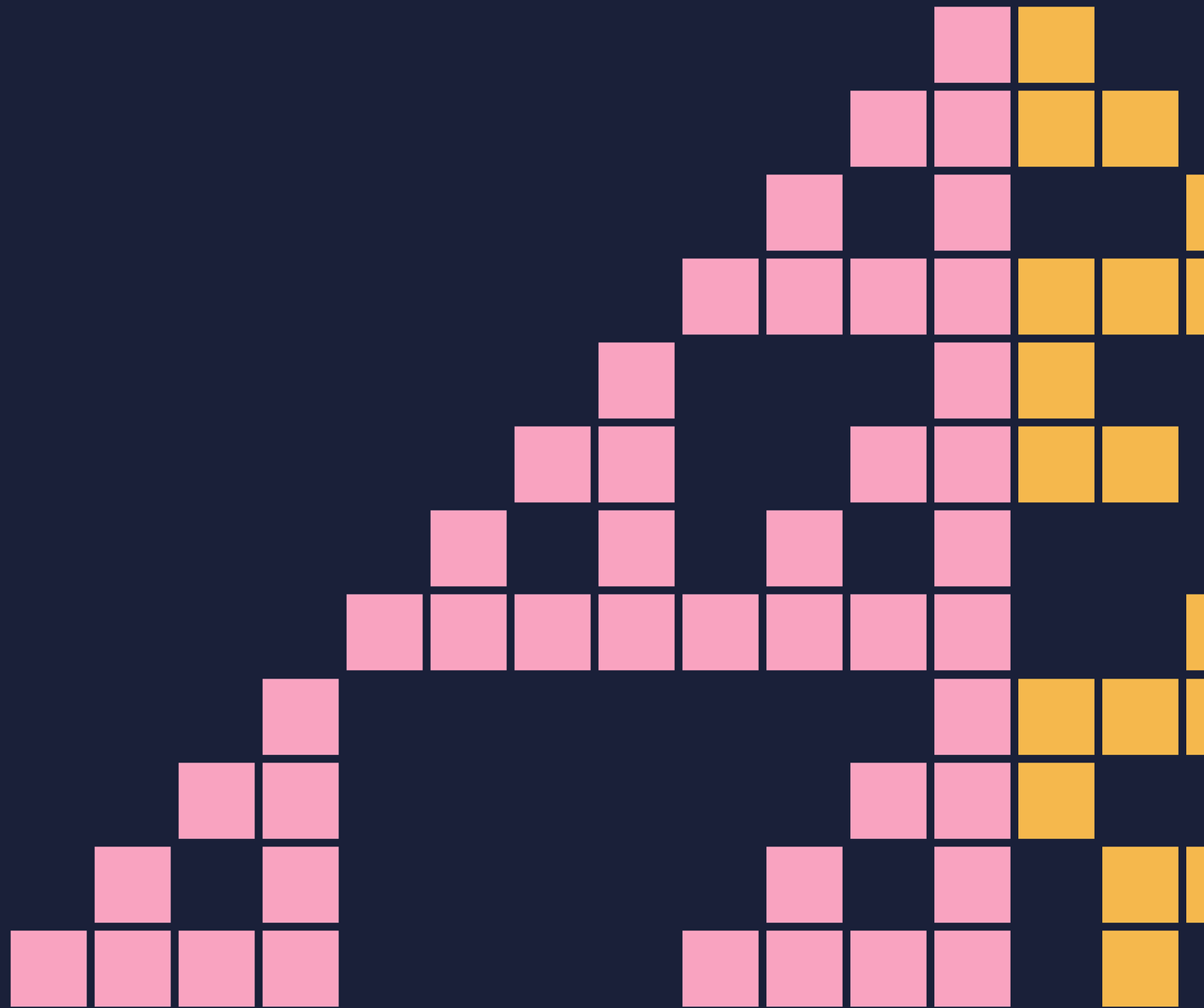
Stage one is narrow: one layer and one platform. Agents do most of the building, and the round pays for a team of five.

Give every agent a seat.

Automaton makes the computer you already own the place where AI works: many agents, your own data, your own machine.

I am looking for co-founders and first investors.

Ronald Djakou
rkouatch@caltech.edu
djakouronald@outlook.com
Demo: youtu.be/Ugkh9cJ_mVg

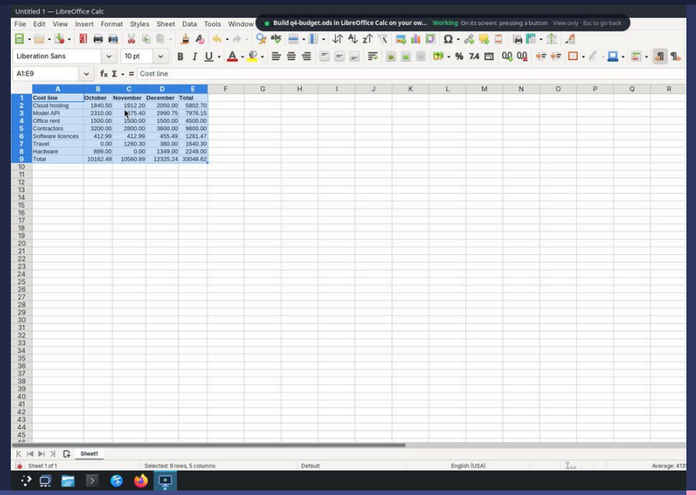


Own desktop, own files, one view.

Automaton works inside the operating system, not as a program on top of it. Four things make an agent a user.

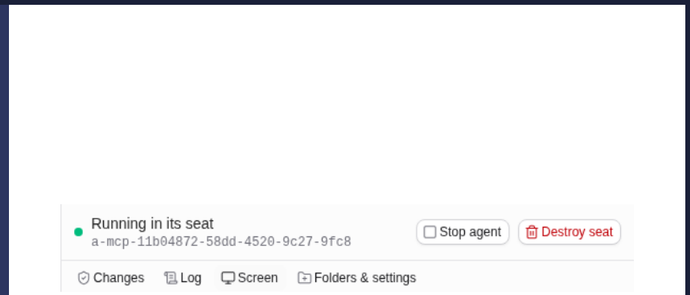
Its own desktop

Every agent gets a private desktop with a screen, a pointer and a keyboard of its own. It opens ordinary apps there, and its clicks never land on your screen.



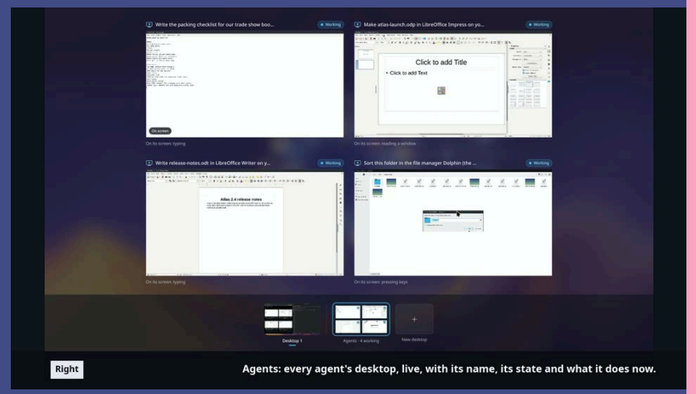
Its own files

The agent sees only the folders you give it. What it writes goes to its own layer first. When it has finished you look at the changes and keep or discard them.



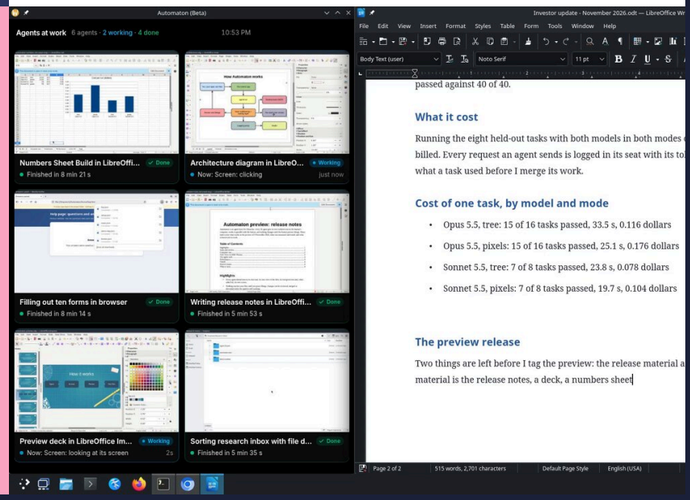
One view of everyone

Swipe up with four fingers, as you would to see your open windows. One card holds every agent's desktop, live, with its state: Working, Done or Needs you.



Your screen stays yours

You keep working while they do. Open any agent's desktop to watch it, take over, or stop it.



What one PC can hold.

Memory in the PC

8 GB	16 GB	32 GB	64 GB

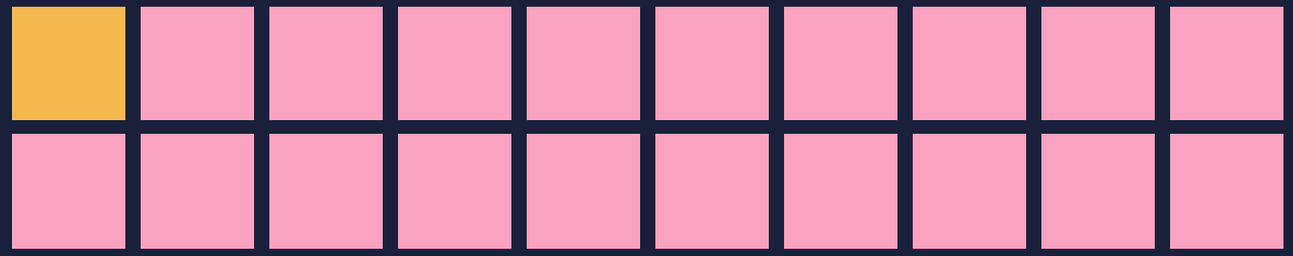
What each agent has open

An office app or a browser	A screen, idle	No screen

- A new agent's screen is ready in 130 milliseconds.
- Three idle agents use 2.2 % of one processor core.
- Six agents with six apps open peaked at 3,975 MB.

19 agents

and you, in a PC with 16 GB of memory



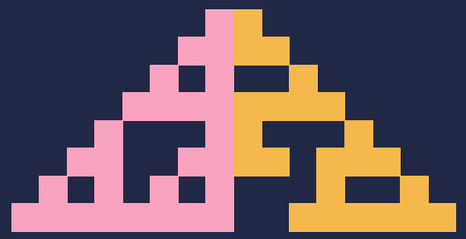
Memory only. 3 GB is set aside for the system and your own apps, and 475 MB for the control app. A working agent also needs processor time and a model; a model that runs on the machine needs memory of its own.

Every number, with its conditions.

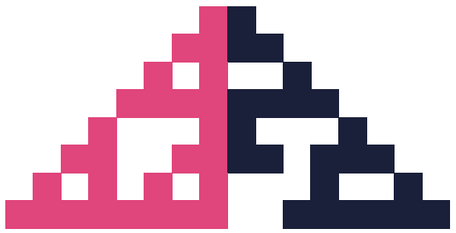
What was measured	Result	Conditions
Three idle agents, panels open	2.2 % of one processor core	11.2 % before tuning. 8 November 2026
Memory for one more agent	221 MB with a screen, 179 MB without	Idle agent
A new agent's screen	Ready in 130 ms	
One seat with a real app	464 to 828 MB at its peak	Six different apps
Six agents with six apps	3,975 MB at the peak	4,741 MB with the control app and the person's apps. 8 GB machine
Six agents, the whole demo	22 min 30 s, all six tasks passed	Sonnet 5.5. $18.09 of model use. Peak 4,955 MB. 9 November 2026
One agent builds a slide deck	7 min 36 s reading the app's structure	17 min 21 s from screenshots. $3.15 against $6.01
Computer use, eight unseen tasks	13.6 to 18.4 s a task, three models	19.6 to 24.4 s from screenshots. 119 of 120 against 120 of 120. 9 November 2026
Windows applications, counted once	46.3 % work, 65.6 % at least in part	697 at random, 13 October 2026. 95 % interval 41.0 to 51.6. Provisional
The same, weighted by installs	24.4 % work, 39.9 % at least in part	95 % interval 23.5 to 25.4
An agent reads a planted secret	17.4 s after it started	105 secret files within reach. Field study, September 2026

The mark: two rules, one peak.

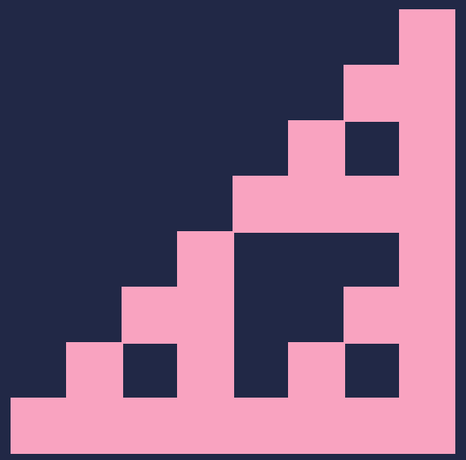
Each half grows from a single cell by a simple rule, one row at a time. Rule 102 on the left makes an orderly pattern that repeats at every scale. Rule 30 on the right never repeats. They meet at the top: two kinds of user, one machine. In this deck pink is the agents and amber is you.



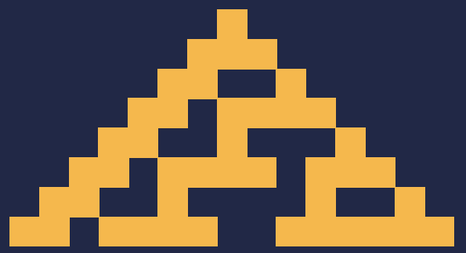
The mark, eight rows. Each slide's foot shows one more row.



On white, for documents



Rule 102 alone: a cell is on when the cell above and its right neighbour differ.



Rule 30 alone: on when the left neighbour differs from "centre or right".

Sources.

- Microsoft, "Experimental agentic features", support.microsoft.com, read 9 November 2026
- Microsoft, "Agentic security" in the Windows security book, learn.microsoft.com, read 9 November 2026
- Automaton, vision document, draft 5
- My handwritten notes, November 2026
- Agent Seat project records: performance, Task View and computer-use findings, 7 to 9 November 2026
- Field study of agents on today's systems, September 2026. Report II (the audit and 11 experiments): kdronald3.github.io/aios-agent-os-pain. Report I (the ranking by cause): the same address with /report1.html
- Windows application sample, 697 units, 13 October 2026
- OSWorld benchmark leaderboard, and Anthropic's announcement of computer use, October 2024
- Epoch AI, "Open models lag state-of-the-art closed models by 4 months", May 2026
- US Census Bureau, Business Trends and Outlook Survey, as reported by the Minneapolis Fed, May 2026
- IDC forecast of 1.3 billion agents by 2028, as quoted by Microsoft
- Cloudera, The Future of Enterprise AI Agents, April 2025. Cisco, Data and Privacy Benchmark Study, January 2026. Deloitte, AI Infrastructure Survey, March 2026
- SteamDB count of Steam Deck Verified and Playable games, September 2026
- Apple, results for fiscal year 2026, and App Store ecosystem study, June 2025. Microsoft, annual report 2026
- Official registry of the Model Context Protocol, count of 8 November 2026. Hugging Face Hub, August 2026
- Carta, State of Pre-Seed, second quarter of 2026
- All outside figures were found by web search on 9 November 2026 and are marked for checking in the notes
- Stephen Wolfram, A New Kind of Science, for Rule 30 and Rule 102